# 1. SOFTWARE PROJECT MANAGEMENT (PhDCS 103)

**MODULE I**

**Overview of Project Management**, PMI Processes, Software project phases, Organizational structures, Project charter, Statement of Work (SOW) **Planning Phase,** Development lifecycle models, Matching lifecycles to projects, Project plans, Work Breakdown Structures (WBS) **Estimation and Budgeting**, Estimation, Budgeting, Project, selection, NPV, ROI, Payback models

**MODULE II**

**Scheduling**, Project network diagram fundamentals, PERT techniques, Gantt charts, Critical chain scheduling **Risk and Change Management**, Mid-term review , Risk management, Change control, More MS-Project.

**Development Management**, Team models, Requirements process, Configuration, management, Software metrics, Programming languages & tools, Managing conflict and motivating, MS-Project: Assigning Resources.

**MODULE III**

**Project Control**, Status reporting, Project metrics, Earned value analysis, Communications Techniques, Process Improvement, MS Project: (a) Resource leveling (b) Other views

**System Test Process,** Test specifications, Black box and white box testing, Test scripts, Unit and integration testing, Acceptance test specifications, Test tools, MS Project:(a) Reporting.

**Final Phases & Other Issues,** Project Recovery, Documentation, Cutover/Migration Post Project Reviews, Closing, MS Project: (a) Advanced features.

## Reference Books:


*1. S. McConnell, "Software Project Survival Guide" (1997)*

*2. S. Berkun, "The Art of Project Management", (2005)*

*3. C. Larman, "Agile and Iterative Development: A Manager's Guide", (2003)*

*4. W. Royce, "Software Project Management: A Unified Framework", (1998)*

*5. J. Highsmith, "Agile Project Management: Creating Innovative Products", (2004)*

*6. T. DeMarco, "The Deadline: A Novel About Project Management", (1997)*

*7. T. DeMarco, "Peopleware: Productive Projects and Teams", (1999)*

8. E. Bennatan, "On Time Within Budget: Software Project Management Practices and Techniques", (2000)

## 2. NETWORKING PROTOCOLS (PhDCS 104)

**MODULE I**

Networks and Services, Approaches to Network Design, The OSI Reference Model; Overview of TCP/IP Architecture, Application Protocols and TCP/IP Utilities, Internet Architecture Interconnection through IP Routers, Internet Protocol (IP), User datagram protocol (UDP).

**MODULE II**

Routing Cores - peers Algorithms Autonomous Systems Exterior Gateway Protocol Multicast Address. Internet Group Management Protocol (IGMP) and Implementation. TCP/IP over ATM networks: ATM cell Transport, Adaptation Layer, IP Address Building in an ATM network Logical IP subnet Concept ATMARP packet format. Domain name system**,** Remote Login (Telnet, Rlogin) File Transfer and Access (FTP, TFTP, NFS), Electronic mail (SMTP, MIME) Internet Management (SNMP, SNMPV2) Internet Security and Firewall Design Post Office Protocol (POP) Network News Transfer Protocol (NNTP).

**MODULE III**

TCP/IP over view- The Transport Layer: TCP and UDP. Elementary TCP Sockets. TCP Client-Server Example. I/O Multiplexing: The select and poll Functions. Socket Options. Elementary UDP Sockets. Elementary Name and Address Conversions. The Client Server Model and Software Design, Concurrent Processing in Client-Server Software, Iterative, Connectionless Servers (UDP), Iterative, Connection-Oriented Servers (TCP), Concurrent, Connection-Oriented Servers (TCP). Single-Process, Concurrent Servers (TCP). Multiprotocol Servers (TCP, UDP), Multiservice Servers (TCP, UDP). Uniform, Efficient Management of server. Concurrency in clients. TCP/IP Architecture, The Internet Protocol, Limitations of IPv4 and Introduction to IPv6, User Datagram Protocol, Transmission Control Protocol, DHCP, Introduction to Internet Routing Protocols

**Reference Books:**

*1. A. Leon-Garcia, Indra Widjaja, "Communication Networks", Tata McGraw Hill, 2000*

*2. William Stallings, "Data and Computer Communications", Pearson Education, 7th Edition.*

*3. Andrew S. Tanenbaum, "Computer Networks", Prentice Hall India, 4th Edition, 2003*

*4. W.Richard Stevens: TCP/IP Illustrated vol 1: The Protocols, Pearson Edun. Asia, 2000.*

*5. Douglas Comer: Internetworking with TCP/IP vol.1: Principles, Protocols and Architecture, Prentice Hall, 4th edition, 2000*

## 3. REAL TIME SYSTES (PhDCS 105)

**Module 1**

Time System, Issues in real time computing, Performance measures of Real Time System, Issues in Real Time Computing, Performance measures of Real time Systems, Real Time Application.Task Assignment and Scheduling: Different task model, Scheduling hierarchy, offline vs Online Scheduling, Clock Drives. Model of Real Time System: Processor, resources, temporal parameter, Periodic Task Model, Sporadic Task Model, Precedence Constraints and Data Dependencies, Scheduling hierarchy Scheduling of Periodic Task: Assumptions, fixed versus dynamic priority algorithms, schedulability test for fixed priority task with arbitrary deadlines.

**MODULE II**

Scheduling of Aperiodic and Sporadic Tasks: Assumptions and approaches, deferrable, sporadic servers, slack stealing in deadline driven and fixed priority systems. Two level scheme for integrated scheduling, Scheduling for applications having flexible constrains.

**MODULE III**

Resources and Resource Access Control: Assumptions on resources and their usage, resource contention, resource access control(Priority Ceiling Protocol, Priority Inheritance protocol, Slack Based Priority Ceiling Protocol, Peremption Ceiling Protocol). Multi Processor Scheduling: Model of multi processor and distributed systems, Scheduling algorithms for end to end periodic tasks in homogeneous/heterogeneous systems, Predictability and validation of dynamic multiprocessor system. Real time Communication: Model of real time Communication, Priority base service For switched network, Weighted Round Robin Service, Medium access Control Protocol, Real Time Protocol.

**Reference Books:**

1. Jane .W. S. Liu Real Time Systems Pearson Education.
2. Krishna .C.M Real Time Systems Mc-Graw Hill Publication